Acids, bases and pH: basic concepts

Virtually all biochemical reactions take place in **aqueous solution**. Therefore, water and its ionization products, **H^+** and **OH^-**, profoundly influence the structure and functions of biomolecules.

Those molecules that interact with the ionization products of water are known as **acids** and **bases**.

Brønsted-Lowry theory for acids and bases

Several theories have been proposed to explain the behavior of acids and bases. A theory that explains well enough the acid-base behavior of biomolecules in water is the Brønsted-Lowry theory. According to this theory, **an acid is a molecule that donates a proton**, and **a base is a molecule that accepts a proton**.

Hydrochloric acid (HCl) is an example of an acid, since in aqueous solution one HCl molecule donates one proton.

The hydroxide ion (OH^-) is a base, since in aqueous solution it accepts a proton.

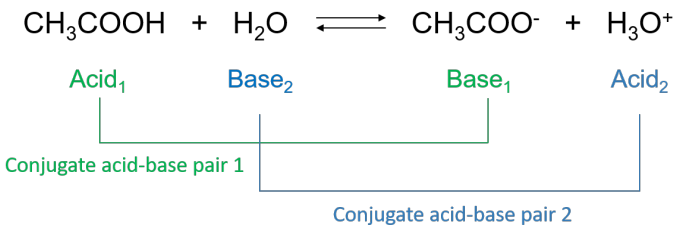
It is important to realize that acids donate protons to other molecules, which accept these protons and, therefore, act as bases. For example, in the case of HCl, the aceptor of H^+ is water:

$$HCl + H_2O \longrightarrow Cl^- + H_3O^+$$

Therefore, free protons (H^+) do not exist as isolated species in water; they are hydrated and are commonly found as hydronium ions *(H_3O^+).*

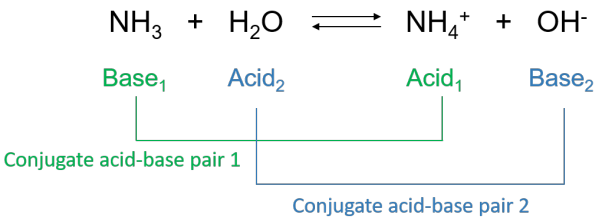
A consequence of the Brønsted-Lowry theory is that in any reaction in which protons are exchanged, two **conjugate acid-base pairs** can be defined.

Let's see the example of acetic acid in aqueous solution:



CH_3COO^- (acetate) is the conjugate base of CH_3COOH (acetic acid) and H_3O^+ is the conjugate acid of H_2O

Another example, in this case of a base:



NH_4^+ (ammonium ion) is the conjugate acid of NH_3 (ammonia) and OH^- is the conjugate base of H_2O

From these two examples, it can be concluded that water is an **amphoteric** molecule; that is, it can act either as an acid or as a base (depending on which other molecule is present in the aqueous solution).

In a practical sense:

- An acid is any substance that increases the concentration of hydrogen ions when dissolved in water.
- A base is any substance that decreases the concentration of hydrogen ions when dissolved in water.

Strength of acids and bases

Not all acid or basic compounds have the same capacity to donate or accept protons:

- A **strong acid** donates protons readily; its reaction with water is essentially complete.
- A **strong base** accepts protons readily; its reaction with water is essentially complete.
- A **weak acid** or a **weak base** establishes an equilibrium in water in which both protonated and deprotonated forms are present.

For example:

Hydrochloric acid is a strong acid and its dissociation in water is nearly complete.

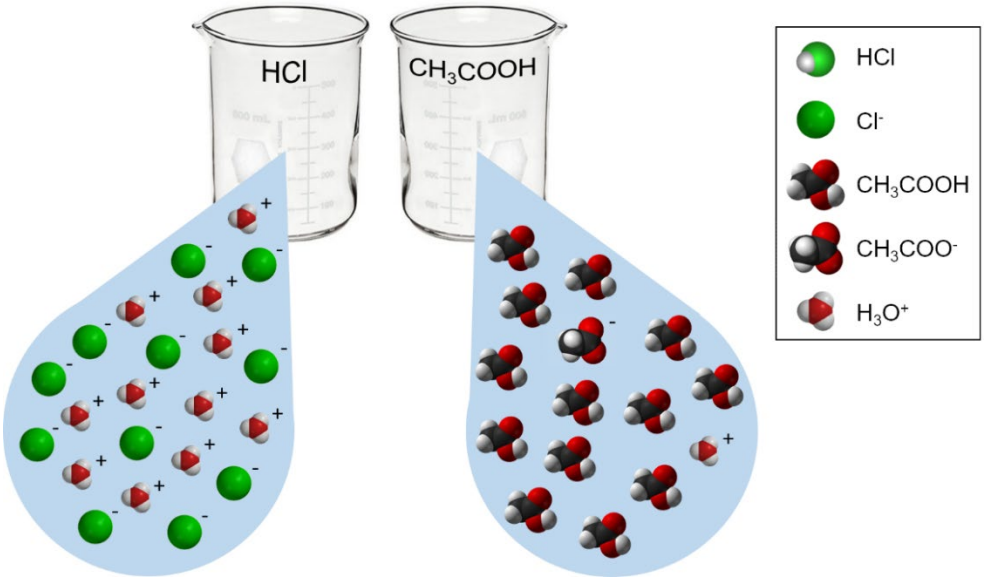
$$HCl + H_2O \longrightarrow Cl^- + H_3O^+$$

(a single left-to-right arrow denotes a complete reaction in that direction)

Acetic acid is a weak acid and only partially dissociates in water.

$$CH_3COOH + H_2O \rightleftarrows CH_3COO^- + H_3O^+$$

(two arrows facing in opposite directions indicate a reversible reaction at equilibrium)



Important: Acid-base functional groups in biological molecules are weak acids or weak bases: carboxyl groups (**-COOH**), amino groups (**-NH_2**), hydroxyl groups (**-OH**), or sulfhydryl (thiol) groups (**-SH**), among others.

The strength of a weak acid is represented with the **acid dissociation constant** (K_a), which is the equilibrium constant of the deprotonation reaction. Therefore, the K_a for a weak acid (AH) dissolved in water ($AH + H_2O \rightleftarrows A^- + H_3O^+$) can be calculated as:

$$K_a = \frac{[A^-] \cdot [H_3O^+]}{[AH]}$$

Note that **the smaller the K_a the weaker the acid**.

In biology, it is common to express the ionization constant as its negative log (**pK**):

$$pK_a = -\log K_a$$

For example, K_a of acetic acid is 1.8×10^{-5}, so its pK_a is 4.74.

In contrast to K_a, **the higher the pK_a the weaker the acid**.

Note that the strength of a weak base can also be described using the pK_a of its conjugate acid.

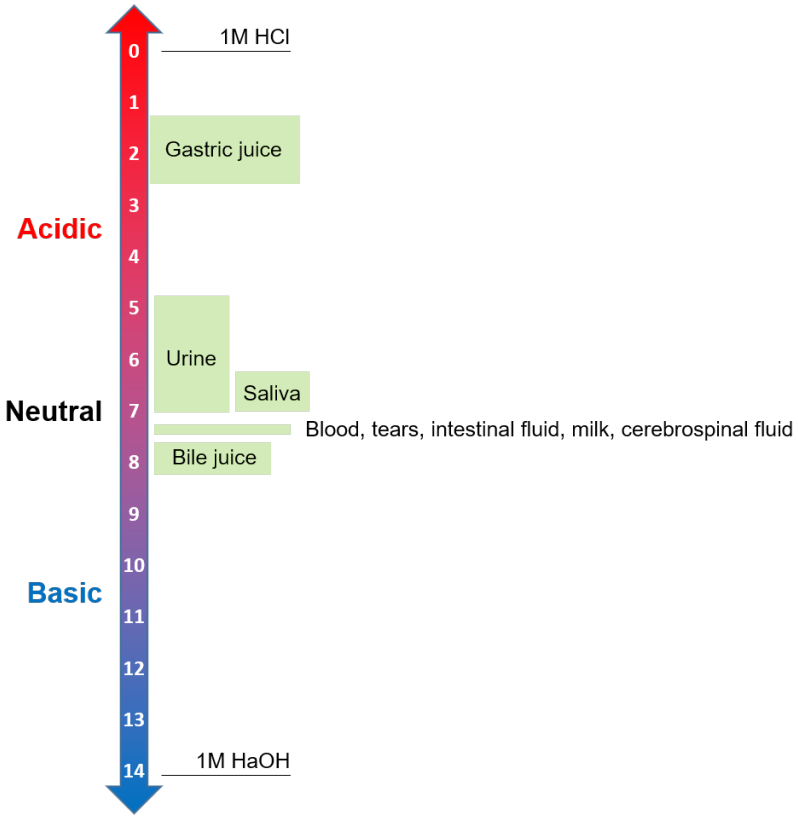
pH

pH is just a convenient way to express the **concentration of hydrogen ions in solution**.

$$pH = -\log[H_3O^+]$$

- A solution with a pH = 7 (i.e. $[H_3O^+] = 10^{-7}$ M) is known as a **neutral solution**. Pure water is a neutral solution because the autoionization of water results in a concentration of H_3O^+ ions of 10^{-7} M.
- A solution with a pH < 7 ($[H_3O^+] > 10^{-7}$ M) is an **acidic solution**.
- A solution with a pH > 7 ($[H_3O^+] < 10^{-7}$ M) is a **basic solution**.

Biological fluids tend to have pH values close to neutral (with some exceptions, such as the gastric juice). The physiological pH range in some fluids is extremely narrow (e.g. for blood between 7.35 and 7.45).



Relationship between a conjugate acid-base pair and the pH of the solution: Henderson-Hasselbalch equation

Looking at the equilibrium of a weak acid in an aqueous solution, one realizes that the amount of the weak acid (e.g. acetic acid) and its conjugate weak base (e.g. acetate) depends on the concentration of hydrogen ions (H_3O^+) in the solution, that is, the pH.

$$AH + H_2O \rightleftarrows A^- + H_3O^+$$

According to Le Chatelier's principle, if the concentration of hydrogen ions is increased in the solution (namely, the pH decreases), the equilibrium will shift to the left. Therefore, the amount of acetic acid molecules will increase and the amount of acetate molecules will decrease.

Alternatively, if the concentration of hydrogen ions is decreased in the solution (namely, the pH increases), the equilibrium will shift to the right. Consequently, the amount of acetic acid molecules will decrease and the amount of acetate molecules will increase.

Knowing the K_a, the proportion of the two forms of the conjugate acid-base pair (e.g. acetic acid/acetate) can be calculated for any given H_3O^+ concentration (and therefore for any given pH):

$$K_a = \frac{[\text{Conjugated base}] \cdot [H_3O^+]}{[\text{Conjugated acid}]}$$

To make it easier, there is a useful way of restating the equation in an expression known as the **Henderson-Hasselbalch equation**:

$$pH = \text{p}K_a + \log\frac{[\text{Conjugate base}]}{[\text{Conjugate acid}]}$$



If you are curious to know how the Henderson-Hasselbalch equation is derived from the K_a equation, watch this video

Example of the practical application of the Henderson-Hasselbalch equation

Problem:

Calculate the concentration of acetic acid and acetate in an aqueous solution at pH 4.26 if the total amount of the acetic acid/acetate pair is 0.2 mM (pK_a of acetic acid = 4.74)

Solution:

$$pH = \text{p}K_a + \log\frac{[\text{Acetate}]}{[\text{Acetic acid}]}$$

$$4.26 = 4.74 + \log\frac{[\text{Acetate}]}{[\text{Acetic acid}]}$$

$$\frac{[\text{Acetate}]}{[\text{Acetic acid}]} = 0.33$$

$$\frac{[\text{Acetate}]}{[0.2 - \text{Acetate}]} = 0.33$$

$$[\text{Acetate}] = 0.05 \text{ mM}$$

$$[\text{Acetic acid}] = 0.15 \text{ mM}$$

Therefore, in any aqueous solution of 0.2 mM acetic acid/acetate at pH 4.26, the concentration of acetic acid is 0.15 mM and the concentration of acetate is 0.05 mM. If the pH changes (by the addition of an acid or a base), the concentrations of the conjugate pair will change accordingly (these new concentrations can be easily recalculated using the Henderson-Hasselbalch equation and the new pH).

Conclusion: The pH determines the ratio of the protonated and deprotonated forms of a weak acid-base pair.

Because protonation can change the electrical charge of biological molecules, changes in pH can profoundly affect their structure, interactions, and function, having important physiological and pathological implications.

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